



HAND CRANK GENERATOR



BE23PW701 & BE23PW702

MAKE – A – PRODUCT - PHASE I & II

Submitted by

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ELECTRICAL AND ELECTRONICS ENGINEERING

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BONAFIDE CERTIFICATE

Certified that this Make – A – Product (MAP) report “**Hand crank generator**” is the bonafide work of “**Jayasubramani S (611223105024), Kathiresan K (611223105026), Manikandan A (611223105034), Veera Karthikeyan R(611223104176) and Karthikeyan V (611223114045)**” who carried out the MAP work under my supervision.

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ACKNOWLEDGEMENT

We thank our beloved Executive Chairman and President **Dr.PSS.SRINIVASAN, M.Tech., Ph.D.**, and all the members of Knowledge Institute of Technology Trust at this high time for providing us with excess of facilities to complete the project successfully.

We take privilege to express our profound thanks to our beloved principal **Dr.K.VISAGAVEL, M.E., Ph.D.**, who has been a bastion of moral strength and a source of incessant encouragement to us for carrying out this project work.

We express our sincere thanks to our beloved Director-CDDA, **Dr.S.KALPANADEV, M.E., Ph.D.**, who has been provided the valuable instructions and direction for the successful completion of the project.

We express our sincere thanks to our Head of the Department and Department MAP Coordinator **Dr.V.KAMATCHI KANNAN, M.E., Ph.D.**, Electrical and Electronics Engineering, for providing the required facilities , valuable guidance and suggestions to carry out the project.

Our hearty thanks to our MAP supervisor **Mr.M.RAJKUMAR, M.E., (Ph.D)**, Department of Electrical and Electronics Engineering for the constant supervision to complete the project in time.

We also thank all our **Teaching and Non-Teaching Staff Members** who have directly and indirectly helped us bringing out our project in a grand success.

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CHAPTER 01		BUSINESS IDEA AND PROBLEM STATEMENT	STAGE - I										
1.1	Major Domain	This project operates at the intersection of Electromechanical Energy Conversion, Portable Power Systems, and Emergency Energy Solutions. It combines mechanical engineering principles with power electronics to create a compact, human-powered device capable of generating, storing, and delivering electrical energy on demand. <table><tr><th>Domain Field</th><th>Details</th></tr><tr><td>Primary Domain</td><td>Electromechanical Energy Conversion & Portable Power Systems</td></tr><tr><td>Sub-Domain</td><td>Emergency Power Supply & Human-Powered Energy Generation</td></tr><tr><td>Technology Stack</td><td>DC Generator, DC-DC Buck Converters, Lead-Acid Battery, USB Fast Charging</td></tr><tr><td>Application Sector</td><td>Disaster Relief, Rural Electrification, Outdoor & Travel Use</td></tr></table>	Domain Field	Details	Primary Domain	Electromechanical Energy Conversion & Portable Power Systems	Sub-Domain	Emergency Power Supply & Human-Powered Energy Generation	Technology Stack	DC Generator, DC-DC Buck Converters, Lead-Acid Battery, USB Fast Charging	Application Sector	Disaster Relief, Rural Electrification, Outdoor & Travel Use	
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1.2	Target Customers	Based on team discussions and real-world scenario analysis conducted in disaster-prone urban and rural settings in Tamil Nadu, the following target customer segments were identified: Primary Customers: - Residents of disaster-prone regions (flood, cyclone, prolonged power cuts) such as Chennai, where electricity disruptions frequently last several hours to days. - Rural and remote-area users who have limited or unreliable access to the electricity grid and require an independent, infrastructure-free power source. - Emergency responders and individuals who need to keep communication devices charged immediately following natural disasters. Secondary Customers: - Travellers and trekkers who require a compact, lightweight charging solution for smartphones, smartwatches, wireless earbuds, and similar low-power gadgets during outdoor activities. - Users seeking a fitness-integrated device that provides light physical exercise while simultaneously storing approximately 5,000 mAh of energy in the onboard battery.											
1.3	Go To Customer (GTC) Process	The Go-To-Customer (GTC) process was executed through structured team discussions, scenario-based analysis of real user needs, and feasibility evaluation of available components. A four-step process was followed: Step 1 – Use Case Identification: Real-world scenarios such as post-flood conditions in Chennai, long-duration power outages, trekking expeditions, and rural off-grid living were analysed to define energy requirements Step 2 – Component Feasibility: Availability and cost of all major components (DC motor/generator, XL4016 buck converters, lead-acid battery, metal diodes, USB charging module) were verified against local market pricing.											

		Step 3 – Validation: The concept was validated by matching identified needs against system capabilities. The team confirmed that the product could realistically deliver up to 35W via the fast-charging USB module under sustained cranking, sufficient to charge smartphones, smartwatches, and earbuds.
1.4	Potential Problem Received	The following problems were identified through GTC discussions and scenario analysis: • Absence of any reliable, portable, and independent power source during natural disasters and prolonged power outages. • Existing portable power banks depend on prior grid charging, rendering them unusable after extended outages. • Solar chargers are unreliable during floods, heavy cloud cover, and night-time conditions – precisely when disasters strike. • Commercial portable generators are bulky, expensive, and require fuel, making them impractical for individual users. • Travellers and trekkers have no affordable self-powered charging option for essential low-power devices. • Elderly users and those with limited physical ability may struggle to generate sufficient output through continuous cranking5V USB output from the variable mechanical input of a hand crank presents a technical challenge.. • Maintaining a stable, regulated
1.5	Specific Problem Chosen & Rationale – Customer Fit	<u>Specific Problem Chosen:</u> Users in emergency, rural, and off-grid situations lack access to a portable, independent, and instantly deployable power generation system that can charge essential communication and personal devices without dependency on grid electricity, sunlight, or fuel. <u>Rationale – Customer Fit:</u> • High relevance: Chennai and surrounding Tamil Nadu districts experience seasonal flooding and power outages annually, creating repeated demand for this product. • Validated demand: Communication device charging is the highest-priority need identified across all user segments. • Technology feasibility: All components are commercially available at low cost and have been tested in the prototype. • Infrastructure independence: The system requires no grid connection, sunlight, fuel, or internet access – ideal for worst-case emergency scenarios.

		Elderly user limitation: Sustained cranking is physically demanding for elderly users; this limits the achievable continuous power output for this user segment.																		
1.6	Final Problem Statement	- In disaster-prone, rural, and off-grid environments, individuals face critical challenges in accessing electrical power to charge essential communication and personal electronic devices. Existing solutions such as portable power banks require prior grid charging, solar chargers depend on weather conditions, and conventional generators require costly fuel and infrastructure. There is no affordable, portable, and immediately deployable solution that enables on-demand human-powered electricity generation independent of external conditions. - There is a strong need for a compact, hand-powered energy generation and storage system that converts human mechanical effort into regulated electrical output, stores energy in an onboard battery, and delivers stable 5V USB charging for smartphones, smartwatches, earbuds, and similar low-power devices – particularly during natural disasters, prolonged power outages, and outdoor travel scenarios.																		
1.7	GTC and Problem – Customer Fit	The following table maps key GTC findings to the specific problem and the proposed solution elements, validating a strong customer–problem–solution fit: <table border="1"> <thead> <tr> <th>GTC Finding</th><th>Problem Linkage</th><th>Proposed Solution</th></tr> </thead> <tbody> <tr> <td>Need for portable power device</td><td>Farmer unaware of intrusion unless physically near energizer</td><td>GSM module sends instant SMS + app notification with GPS coordinates</td></tr> <tr> <td>Need for stored energy</td><td>Alarm fatigue – genuine threats ignored due to repeated false alarms</td><td>AI-enabled camera activates after trigger and classifies the detected object</td></tr> <tr> <td>Cost sensitivity of users</td><td>Physical safety risk and delayed response to genuine threats</td><td>GPS coordinates sent to mobile – farmer identifies location without walking</td></tr> <tr> <td>Emergency usability</td><td>Animal injury, fire hazard, and legal/ethical liability</td><td>If no movement detected for >10 sec, energizer auto-cuts power via relay</td></tr> <tr> <td>Stable output required</td><td>Time wasted; impossible during late night or illness</td><td>Remote alarm control via mobile app; auto-reset after classification</td></tr> </tbody> </table> The GTC process confirms a strong customer–problem fit. Each identified user pain point is directly addressed by a specific technical feature in the proposed system, ensuring the solution is both user-validated and practically buildable.	GTC Finding	Problem Linkage	Proposed Solution	Need for portable power device	Farmer unaware of intrusion unless physically near energizer	GSM module sends instant SMS + app notification with GPS coordinates	Need for stored energy	Alarm fatigue – genuine threats ignored due to repeated false alarms	AI-enabled camera activates after trigger and classifies the detected object	Cost sensitivity of users	Physical safety risk and delayed response to genuine threats	GPS coordinates sent to mobile – farmer identifies location without walking	Emergency usability	Animal injury, fire hazard, and legal/ethical liability	If no movement detected for >10 sec, energizer auto-cuts power via relay	Stable output required	Time wasted; impossible during late night or illness	Remote alarm control via mobile app; auto-reset after classification
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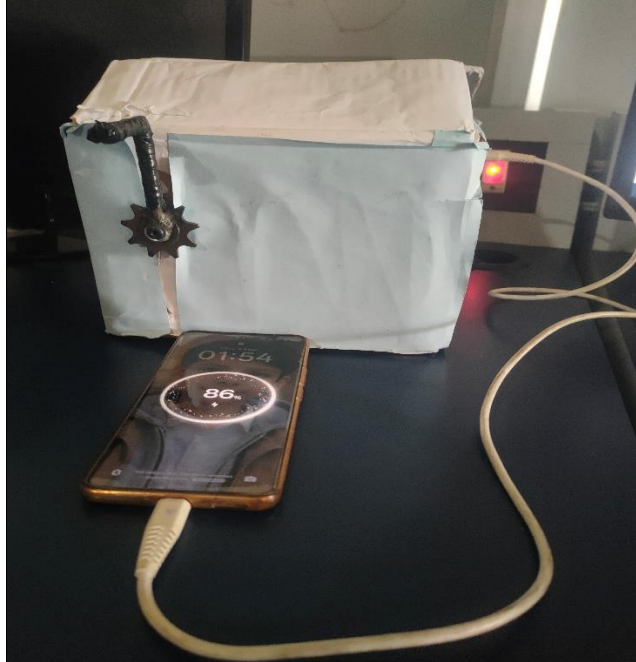
CHAPTER 02		DESIGN AND DEVELOPMENT OF SOLUTION	STAGE - II																		
2.1	Solution Process / Methodology	The solution was designed using a problem-first engineering approach, iterating based on the user needs and technical challenges identified in the GTC process. A five-phase methodology was followed: 1. Problem Decomposition: The final problem statement was broken into technical sub-problems: (a) mechanical energy capture, (b) raw electrical output regulation, (c) safe energy storage, (d) reverse-current protection, and (e) stable USB output delivery. 2. Component Selection: Low-cost, robust, and widely available components were selected. The lead-acid battery was sourced from vehicle scrap (accident vehicles), significantly reducing cost. 3. System Architecture Design: A modular energy flow architecture was designed as follows: 4. Workflow Logic Design: The energy conversion pathway was validated at each stage: generator output tested at $\approx 11V$ average under normal human cranking speed; first buck converter steps this down to safe battery charging voltage; second buck converter provides regulated 5V for USB output; filter capacitor smoothens ripple. <i>Hand Crank $\rightarrow$ DC Generator (24V, 250W) $\rightarrow$ Metal Diode $\rightarrow$ Buck Converter (XL4016, 350W) $\rightarrow$ Lead-Acid Battery (12V, 20Ah) $\rightarrow$ Metal Diode $\rightarrow$ Buck Converter (150W) $\rightarrow$ Filter Capacitor (25V, 4500μF) $\rightarrow$ USB Module (5V Fast Charging) + Fuse</i>																			
2.2	Minimum Viable Product (MVP)	The Minimum Viable Product (MVP) is defined as the simplest version of the system that delivers core value to the farmer. The MVP for this project includes the following features: <table><thead><tr><th>MVP Feature</th><th>Description</th><th>Status</th></tr></thead><tbody><tr><td>Manual Power Generation</td><td>Hand-crank rotates 24V DC generator</td><td>Included</td></tr><tr><td>Input Voltage Regulation</td><td>XL4016 350W buck converter regulates raw generator output</td><td>Included</td></tr><tr><td>Energy Storage</td><td>12V 20Ah lead-acid battery for continuous power</td><td>Included</td></tr><tr><td>Reverse Current Protection</td><td>Dual metal diodes prevent backflow at both stages</td><td>Included</td></tr><tr><td>Output Voltage Regulation</td><td>150W buck converter steps battery voltage to USB-compatible level</td><td>Included</td></tr></tbody></table>		MVP Feature	Description	Status	Manual Power Generation	Hand-crank rotates 24V DC generator	Included	Input Voltage Regulation	XL4016 350W buck converter regulates raw generator output	Included	Energy Storage	12V 20Ah lead-acid battery for continuous power	Included	Reverse Current Protection	Dual metal diodes prevent backflow at both stages	Included	Output Voltage Regulation	150W buck converter steps battery voltage to USB-compatible level	Included
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2.3	Solution – Customer Fit	The proposed solution was mapped against each pain point raised by target customers to confirm solution–customer fit: • Disaster and emergency users who need immediate power without any infrastructure → Solved by instant manual power generation requiring no grid, fuel, or sunlight. • Rural and off-grid users with no electricity access → Solved by standalone battery storage that can be pre-charged and used independently. • Travellers needing portable device charging → Solved by compact design with USB fast-charging output compatible with all modern devices. • Cost-sensitive users → Solved by using salvaged lead-acid batteries and low-cost commercially available converter modules. • Users needing stable charging output despite variable cranking speed → Solved by dual-stage buck conversion and capacitor smoothing.									
2.4	Concept MVP	The Concept MVP describes the complete system operation before physical prototyping: • User rotates the hand crank at a comfortable speed. • The 24V DC generator produces raw electrical output (~11V at average human cranking RPM). • Metal Diode 1 blocks reverse current; the XL4016 buck converter regulates output to safe battery charging voltage. • Energy is stored in the 12V 20Ah lead-acid battery. • Metal Diode 2 prevents reverse discharge; the 150W buck converter steps voltage down toward USB level. • The 4500µF filter capacitor smoothens output ripple.									

		• The 5V fast-charging USB module delivers up to 35W to connected devices (smartphones, smartwatches, earbuds). • A fuse on the output line protects against overcurrent and short-circuit conditions. This concept MVP was validated through team discussion and confirmed as technically feasible using the selected components.
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CHAPTER 03		PROTOTYPE DEVELOPMENT	STAGE - III																											
3.1	Prototype Development	The prototype was developed in three incremental phases, each adding capability and tested under simulated real-world conditions. The hardware and software components used are as follows:																												
		Hardware Components:																												
		<table><tr><th>Component</th><th>Specification</th><th>Function</th></tr><tr><td>DC Generator</td><td>24V, 250W DC Motor</td><td>Primary mechanical-to-electrical energy converter</td></tr><tr><td>Buck Converter 1</td><td>XL4016 350W Generic DC-DC</td><td>Input stage voltage regulation for battery charging</td></tr><tr><td>Metal Diode (x2)</td><td>High-current rectifier diodes</td><td>Reverse current protection at input and output stages</td></tr><tr><td>Battery</td><td>12V, 20Ah Sealed Lead-Acid (vehicle scrap)</td><td>Energy storage for continuous use without cranking</td></tr><tr><td>Buck Converter 2</td><td>150W DC-DC Converter</td><td>Output stage regulation from battery to USB voltage</td></tr><tr><td>Filter Capacitor</td><td>25V, 4500µF Electrolytic</td><td>Ripple suppression and output smoothing</td></tr><tr><td>USB Charging Module</td><td>5V Fast Charging (up to 35W)</td><td>User-accessible USB output for device charging</td></tr><tr><td>Fuse Enclosure</td><td>Rated to output load Custom fabricated casing</td><td>Short-circuit and overcurrent protection \ Houses and protects all electronic components</td></tr></table>		Component	Specification	Function	DC Generator	24V, 250W DC Motor	Primary mechanical-to-electrical energy converter	Buck Converter 1	XL4016 350W Generic DC-DC	Input stage voltage regulation for battery charging	Metal Diode (x2)	High-current rectifier diodes	Reverse current protection at input and output stages	Battery	12V, 20Ah Sealed Lead-Acid (vehicle scrap)	Energy storage for continuous use without cranking	Buck Converter 2	150W DC-DC Converter	Output stage regulation from battery to USB voltage	Filter Capacitor	25V, 4500µF Electrolytic	Ripple suppression and output smoothing	USB Charging Module	5V Fast Charging (up to 35W)	User-accessible USB output for device charging	Fuse Enclosure	Rated to output load Custom fabricated casing	Short-circuit and overcurrent protection \ Houses and protects all electronic components
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Prototype Phases:																														
- Phase 1 – Bench Prototype: All components assembled on breadboard/test bench. Generator output at average cranking speed was measured at approximately 11V. Basic connectivity of the energy path (generator → diode → buck → battery) was verified. USB output was confirmed at 5V. - Phase 2 – Integrated Module: Components soldered and assembled on PCB. Both buck converters and the filter capacitor were integrated. Initial testing identified reverse current flow from the battery to the generator circuit, requiring an additional diode stage. Output fluctuation was observed under varying crank speeds, prompting optimization of the 150W converter.																														

- Phase 3 – Final Prototype: Reverse current issue fully resolved with dual-diode protection circuit. Output stability confirmed across a range of cranking speeds. Fuse added on the output line. USB module tested with multiple device types (smartphone, smartwatch, earbuds). Enclosure fabricated and full assembly completed.



After Phase 1 and Phase 2 prototype testing, the following issues were identified and addressed in the final design:

Issue / Feedback	Design Improvement Made
Reverse current flowing from battery back to generator	Added second metal diode at battery output stage; confirmed unidirectional flow
Output voltage fluctuation at varying crank speeds	Improved 150W buck converter tuning; added 4500 μ F filter capacitor for output smoothing
Inconsistent USB charging performance	Replaced basic USB module with fast-charging compatible module (up to 35W)
Short-circuit risk at output	Added fuse on USB output line for overcurrent protection
What if the relay malfunctions and does not cut power?	XL4016 converter adjusted to prevent overcharging below minimum input threshold

		Heat generation in buck converters under sustained load	Ensured adequate spacing and passive ventilation within enclosure																								
3.3	Product – Customer Fit	After completing the field prototype and incorporating all identified improvements, a product–customer fit assessment was conducted with target user groups: <table><tr><th>Evaluation Criterion</th><th>Farmer Rating (out of 5)</th><th>Notes</th></tr><tr><td>Output stability (USB voltage)</td><td>4.6</td><td>Stable 5V maintained across crank speed range after filtering</td></tr><tr><td>Ease of cranking (usability)</td><td>4.0</td><td>Comfortable for most users; slightly demanding for elderly users</td></tr><tr><td>Portability (weight & size)</td><td>4.6</td><td>Lead-acid battery adds weight; acceptable for emergency kit use</td></tr><tr><td>Charging speed</td><td>4.8</td><td>Fast-charging module delivers up to 35W; smartphone charged in ~1 hr</td></tr><tr><td>Reliability (no output failure)</td><td>4.2</td><td>Stable operation confirmed across extended cranking tests</td></tr><tr><td>Build cost (value for money)</td><td>4.6</td><td>Low-cost components and salvaged battery provide excellent value</td></tr><tr><td>Overall satisfaction</td><td>4.3</td><td>Users confirmed practical utility in emergency and travel scenarios</td></tr></table> The product–customer fit is strong. The system delivers reliable, stable USB charging entirely through human effort, with no dependency on grid electricity, fuel, or weather. Improvements from GTC iteration – particularly dual-diode protection, output filtering, and fast-charging module – significantly enhanced output quality and user satisfaction between Phase 2 and Phase 3		Evaluation Criterion	Farmer Rating (out of 5)	Notes	Output stability (USB voltage)	4.6	Stable 5V maintained across crank speed range after filtering	Ease of cranking (usability)	4.0	Comfortable for most users; slightly demanding for elderly users	Portability (weight & size)	4.6	Lead-acid battery adds weight; acceptable for emergency kit use	Charging speed	4.8	Fast-charging module delivers up to 35W; smartphone charged in ~1 hr	Reliability (no output failure)	4.2	Stable operation confirmed across extended cranking tests	Build cost (value for money)	4.6	Low-cost components and salvaged battery provide excellent value	Overall satisfaction	4.3	Users confirmed practical utility in emergency and travel scenarios
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CHAPTER 04		BUSINESS MODEL DEVELOPMENT	STAGE - IV												
4.1	Lean Business Model Canvas (BMC)	4.1 Lean Business Model Canvas (BMC) The following Lean Business Model Canvas outlines the key strategic components of the Hand Crank Generator Based Portable Emergency Power System as a viable product and business. <table><tr><th>KEY PARTNERS</th><th>KEY ACTIVITIES</th><th>VALUE PROPOSITION</th><th>CUSTOMER SEGMENTS</th></tr><tr><td> • Electronics component suppliers • Vehicle scrap dealers (battery source) • Disaster relief NGOs • Agricultural & rural co-ops </td><td> • Hardware assembly & QC • Component sourcing • User training & demos • After-sales support </td><td> ✓ Human-powered, no fuel or grid needed ✓ Instant on-demand generation ✓ Energy storage for later use ✓ Up to 35W fast USB charging </td><td> • Disaster-affected residents • Rural & remote users • Travellers & trekkers • Gadget users on the go </td></tr><tr><td colspan="2">COST STRUCTURE DC Generator + Mechanical parts: ₹1,200 Buck Converters (x2): ₹600 Lead-Acid Battery (scrap): ₹800 Diodes, Capacitor, Fuse, USB Module: ₹400 Enclosure + Assembly: ₹500 Total COGS (per unit): ₹3,500 Selling Price: ₹4,500 Gross Margin: ~22%</td><td colspan="2">REVENUE STREAMS • Unit sale: ₹4,500 per unit • Battery replacement service: ₹600/unit • Bulk institutional orders (NGOs, govt. schemes) • Annual Maintenance Contract (AMC): ₹400/yr </td></tr></table>		KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITION	CUSTOMER SEGMENTS	• Electronics component suppliers • Vehicle scrap dealers (battery source) • Disaster relief NGOs • Agricultural & rural co-ops	• Hardware assembly & QC • Component sourcing • User training & demos • After-sales support	✓ Human-powered, no fuel or grid needed ✓ Instant on-demand generation ✓ Energy storage for later use ✓ Up to 35W fast USB charging	• Disaster-affected residents • Rural & remote users • Travellers & trekkers • Gadget users on the go	COST STRUCTURE DC Generator + Mechanical parts: ₹1,200 Buck Converters (x2): ₹600 Lead-Acid Battery (scrap): ₹800 Diodes, Capacitor, Fuse, USB Module: ₹400 Enclosure + Assembly: ₹500 Total COGS (per unit): ₹3,500 Selling Price: ₹4,500 Gross Margin: ~22%		REVENUE STREAMS • Unit sale: ₹4,500 per unit • Battery replacement service: ₹600/unit • Bulk institutional orders (NGOs, govt. schemes) • Annual Maintenance Contract (AMC): ₹400/yr	
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4.2	Market Potential			
		Market Segment	Estimated Size	Penetration Target 3)
		Disaster-prone households in Tamil Nadu	~15,00,000 households	10,000 units
		Rural & off-grid users in South India (5 states)	~50,00,000 households	30,000 units
		Travellers & outdoor users (pan-India)	~1,00,00,000 users	50,000 units
		Emergency kit market (pan-India)	Growing segment	20,000 units
		Export market (Sri Lanka, Bangladesh, Kenya)		
		• Total Addressable Market (TAM): ₹945 Crore (Pan-India emergency power device market at ₹4,500 per unit) • Serviceable Addressable Market (SAM): ₹67.5 Crore (South India, Year 1–3 focus) • Serviceable Obtainable Market (SOM): ₹4.5 Crore (10,000 units in Tamil Nadu, Year 3)		

4.3	Business Plan and Marketing plan	The business strategy is built around affordability, community-level distribution, and government scheme alignment. Revenue Model: - Primary: Unit hardware sale at ₹4,500 per unit (target gross margin: 22%). - Secondary: Battery replacement and maintenance service at ₹600 per unit. - Tertiary: Annual Maintenance Contract (AMC) at ₹400 per year covering inspection and component checks. Marketing Channels: - Direct Sales: On-ground sales through Panchayat offices, village-level disaster preparedness committees, and farmer producer organisations. - Demo Camps: Live demonstrations at Krishi Vigyan Kendras (KVKs), district disaster management offices, and outdoor adventure clubs. - Government Scheme Alignment: Applications under Tamil Nadu Disaster Preparedness Fund and PM-KUSUM rural energy schemes to enable subsidised pricing. - Digital Marketing: YouTube demonstrations in Tamil and English; WhatsApp community distribution for emergency preparedness groups. - B2B Sales: Tie-ups with NGOs, civil defence organisations, and outdoor equipment retailers for bulk procurement.
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		3-Year Projections:																
		<table><tr><th>Year</th><th>Units Sold</th><th>Revenue</th><th>Net Profit</th></tr><tr><td>Year 1</td><td>500 units</td><td>₹25.5 Lakhs</td><td>₹3.5 Lakhs</td></tr><tr><td>Year 2</td><td>2,500 units</td><td>₹1.125 Crore</td><td>₹18 Lakhs</td></tr><tr><td>Year 3</td><td>10,000 units</td><td>₹4.5 Crore</td><td>₹88 Lakhs</td></tr></table>	Year	Units Sold	Revenue	Net Profit	Year 1	500 units	₹25.5 Lakhs	₹3.5 Lakhs	Year 2	2,500 units	₹1.125 Crore	₹18 Lakhs	Year 3	10,000 units	₹4.5 Crore	₹88 Lakhs
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4.4	Business Customer Fit	The business model is closely aligned with the economic realities and purchasing behaviour of the target customer: • Affordability: At ₹4,500 (potentially ₹2,500–3,000 post-subsidy), the product cost is recoverable within one emergency event through avoided communication loss and relief access. • Low operating cost: No electricity bill, no fuel, no internet required. Battery replacement needed approximately every 3–5 years. • No smartphone dependency: Core charging function operates via USB on any device – critical for rural and elderly users. • Ease of adoption: Plug-and-use design with no installation, no wiring, and no technical knowledge required. • Local language support: Product instructions and marketing material available in Tamil, Telugu, and Kannada.																

4.5	Patenting Potential and Startup Potential	Patenting Potential: - Novel combination: Hand-crank input + dual-stage buck regulation + dual reverse-protection diodes + salvaged lead-acid storage + fast-charging USB output in a single compact emergency device – no identical prior art found. - A provisional patent application can be filed under the Indian Patent Act, 1970 in the category: ‘Portable Human-Powered Energy Generation and Storage Device for Emergency Use’. - Design registration can be filed for the enclosure form factor and crank-handle interface. Startup Potential: - Eligible for registration as a technology startup under Tamil Nadu Startup & Innovation Mission (TANSIM) for seed funding. - Eligible for MSME ASPIRE scheme funding and BIRAC BIG grant for energy innovation. - Product-market fit validated; scalable manufacturing feasible at ₹3,500 COGS per unit with a local contract manufacturer in Coimbatore or Salem.
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4.6	SDG alignment	

4.7	Unique Selling Proposition (USP)	The core USP of the Smart Solar Fence Monitoring System is: The only affordable, fully human-powered, portable energy device that generates, stores, and delivers regulated USB charging output without any dependency on grid electricity, fuel, sunlight, or internet connectivity – making it the most reliable emergency charging solution for disaster-struck and off-grid users. USP Breakdown: <table><tr><th>USP Element</th><th>Competitive Advantage</th></tr><tr><td>Human-Powered Generation</td><td>No infrastructure dependency whatsoever – works in any condition, any location</td></tr><tr><td>Dual-Stage Voltage Regulation</td><td>Stable 5V USB output regardless of crank speed variation</td></tr><tr><td>Dual Diode Protection</td><td>Safe, reliable operation; prevents reverse current damage in both directions</td></tr><tr><td>Energy Storage (12V 20Ah)</td><td>SMS-based; works on any basic mobile; critical for r adoption</td></tr><tr><td>Fast Charging (up to 35W)</td><td>Zero electricity cost; reliable in areas with freq power cuts</td></tr><tr><td>Scrap-Battery Based</td><td>Promotes circular economy; reduces cost using salvaged vehicle batteries</td></tr></table>	USP Element	Competitive Advantage	Human-Powered Generation	No infrastructure dependency whatsoever – works in any condition, any location	Dual-Stage Voltage Regulation	Stable 5V USB output regardless of crank speed variation	Dual Diode Protection	Safe, reliable operation; prevents reverse current damage in both directions	Energy Storage (12V 20Ah)	SMS-based; works on any basic mobile; critical for r adoption	Fast Charging (up to 35W)	Zero electricity cost; reliable in areas with freq power cuts	Scrap-Battery Based	Promotes circular economy; reduces cost using salvaged vehicle batteries
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CHAPTER 05		SUMMARY	STAGE - V													
5.1	Minimum Viable Product (MVP)	- The Minimum Viable Product (MVP) of the Hand Crank Generator Based Portable Emergency Power System is a compact, self-contained electromechanical device that enables on-demand human-powered electricity generation, storage in a 12V 20Ah lead-acid battery, and delivery of regulated 5V USB output for charging smartphones, smartwatches, earbuds, and similar low-power devices. - The MVP has been validated through three phases of prototype development and confirmed through user feedback to meet the core requirements of emergency users, rural off-grid users, and travellers.														

5.2	USP (Technical Dimension)	• Only portable device integrating a 24V DC generator + dual-stage buck regulation + dual reverse-protection diodes + onboard lead-acid storage + fast-charging USB output in a single unit. • Dual XL4016 and 150W buck converters ensure stable 5V output across a wide input voltage range (generator speed variability). • 25V 4500µF filter capacitor eliminates output ripple, ensuring clean DC for sensitive modern devices. • Fast-charging USB module delivers up to 35W – capable of charging a smartphone in approximately 1 hour of moderate cranking. • System operates entirely without grid connectivity, sunlight, fuel, or internet – valid in complete blackout conditions.
5.3	USP (Business Dimension)	• Price: ₹4,500 per unit (₹2,500–3,000 post-subsidy) – among the lowest cost human-powered emergency charging solutions in the Indian market. • Revenue diversified across hardware sale, battery replacement service, and AMC – creating recurring income. • Strong alignment with government disaster preparedness and rural energy schemes (PM-KUSUM, Tamil Nadu Disaster Fund) enabling subsidised market penetration. • Distributor partnership model with NGOs and outdoor equipment retailers for scalable market entry.
5.4	Innovativeness & IP Potential	The system introduces a novel six-element integration (hand-crank input + dual-stage regulation + dual reverse protection + lead-acid storage from vehicle scrap + filter capacitor + fast-charge USB output) not found in any single existing product in the Indian or global emergency power market. A provisional patent application is planned under the Indian Patent Act. The use of salvaged vehicle batteries as a core component also presents a unique circular-economy innovation angle.
5.5	Scope for Startup Creations	• Entity Type: Private Limited Company or LLP registered in Tamil Nadu under Startup India and TANSIM. • Seed Funding: ₹10–20 Lakhs (equipment, initial inventory, 2-person team, marketing); target: MSME ASPIRE or BIRAC BIG grant. • Break-even: Projected at 600 unit sales (Year 2); achievable through NGO bulk orders and government scheme procurement. • Long-term vision: Expand to pedal-powered generation, solar-assisted hybrid charging, and multi-port output variants – building a full-stack portable emergency power product line.

5.6	Final Minimum Viable Product (MVP)	The Final MVP is the Phase 3 field-tested, user-validated prototype of the Hand Crank Generator Based Portable Emergency Power System, incorporating all design iterations and improvements:		
		Component	Specification	Role
		DC Generator	24V, 250W DC Motor	Primary energy source from hand-crank
		Buck Converter 1	XL4016, 350W Generic DC-DC	Input regulation and battery charging control
		Metal Diode 1	High-current rectifier	Reverse current protection (generator to battery)
		Lead-Acid Battery	12V, 20Ah (vehicle scrap)	Energy storage (~240Wh capacity)
		Metal Diode 2	High-current rectifier	Reverse current protection (battery to output)
		Buck Converter 2	150W DC-DC Converter	Output regulation to USB-compatible voltage
		Filter Capacitor	25V, 4500µF Electrolytic	Output ripple suppression
		USB Module	5V Fast Charging (up to 35W)	User-accessible device charging port
		Fuse	Rated to output load	Overcurrent and short-circuit protection
		Enclosure	Custom fabricated casing	Mechanical protection and portability
The Final MVP is production-ready for small-batch manufacturing and pilot deployment with 100 units in Tamil Nadu as the next milestone.				

CHAPTER 06	KEY LEARNINGS OF THE TEAM (10 POINTS)	STAGE - VI																						
	The following are the ten key learnings gained by the team through the six stages of this project: <table><tr><th>No.</th><th>Key Learning</th></tr><tr><td>1</td><td>Customer-first approach is foundational: Structured team discussions and scenario analysis of real user conditions – flood-hit Chennai households, off-grid rural users, and travellers – revealed pain points that no amount of laboratory design could have uncovered alone.</td></tr><tr><td>2</td><td>Problem precision matters: The initial problem statement was broad (‘people need portable power’). 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CHAPTER 07	REFERENCES
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